

SAI TEJA MEKA

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PROFESSIONAL EXPERIENCE

AI Engineer Intern | Cloud Bridge Solutions Inc, Marlborough, MA

Sep 2025 – Present

- Architected enterprise-scale AI education platform with plugin-based microservices, 9 production-ready modules (RAG-powered tutor, adaptive testing, lesson generation, analytics) across 15,000+ lines of Python/JavaScript using Flask Blueprint and MongoDB with 18+ collections
- Designed RAG engine with sub-250ms semantic retrieval using Pinecone, document chunking pipeline processing 500+ documents with OpenAI embeddings, achieving 92%+ relevance across 1,200+ query scenarios
- Engineered AI content generation services with GPT-4 API, lesson planner with curriculum alignment, adaptive quiz system with difficulty progression, and automated grading, achieving 88% accuracy against teacher-validated rubrics

Data Science Consultant | PepsiCo Capstone Graduate Project, Valhalla, NY

Aug 2024 – Dec 2024

- Developed predictive ML models (linear regression, gradient-boosted trees) optimizing packaging performance with feature engineering on material properties, improving prediction accuracy across multi-variable datasets
- Delivered a simulation tool enabling researchers to test packaging recipe outcomes via automated virtual experimentation, accelerating R&D decision velocity by 25%

Data Analyst | Savant Instruments Pvt. Ltd, Hyderabad, India

Jul 2022 – Jul 2023

- Engineered real-time Power BI dashboards and AI-powered reporting pipelines for B2B water analytics, reducing manual reporting overhead by 40%
- Automated Python/SQL ETL workflows and built forecasting engines, improving prediction accuracy by 25% with end-to-end experiment tracking

SKILLS

- AI/ML Engineering:** Multi-agent systems, LLM orchestration, RAG pipelines, LoRA fine-tuning (Unsloth), vLLM serving, Prompt engineering, Vector databases (pgvector, ChromaDB, Pinecone), Knowledge graphs (Neo4j), Model evaluation, Embeddings, Semantic search, GRPO/LADDER reinforcement learning, Claude Code, Ollama, LM Studio
- Backend & APIs:** Python, FastAPI, Node.js, REST APIs, WebSockets, JWT authentication, PostgreSQL, Neo4j, MongoDB, Event sourcing, Real-time streaming, Async processing
- Frontend/Visualization:** React, Next.js, TypeScript, React Three Fiber, Streamlit, React Flow, Tailwind CSS
- Data & Infrastructure:** PyTorch, LangChain, Crew AI, OpenAI/Anthropic APIs, Docker, Git, SQL, NumPy, Scikit-learn, LM Studio, RunPod, LiteLLM
- Cloud & DevOps:** AWS (EC2, S3, Lambda), Google Cloud Platform, RunPod (GPU serving), Hugging Face Hub, CI/CD, Container orchestration LangChain, CrewAI, Hugging Face

PROJECT EXPERIENCE

Combo (Cognitive Lens Engine) - Multi-Persona LLM Platform

(PostgreSQL • pgvector • Neo4j • vLLM • LoRA/Unsloth • FastAPI • React • RunPod • LiteLLM)

- Architected a multi-persona AI platform where each specialist is individually fine-tuned with its own reasoning style, Einstein persona (LLaMA 3.1 8B, 1,520 training examples), and Math specialist (Qwen2.5-Math-7B) deployed via vLLM with native LoRA adapter switching on RunPod A40 GPUs
- Built a semantic claims extraction pipeline (PostgreSQL + pgvector + Neo4j) that captures cognitive DNA from every response, metabolizes it back into the persona's body, and enables autonomous self-evolution, a living loop where the system learns from its own outputs
- Engineered soul.py routing layer with intelligent fallback (fine-tuned model → Claude API), two-tier prompting (static bio anchor + pgvector semantic retrieval), and a slim system prompt architecture where identity and knowledge are baked into weights rather than context
- Completed LADDER reinforcement learning cycle for math specialist: SFT → 79%, GRPO → 72% (catastrophic forgetting), LADDER+replay buffer → 78% recovery, proving the full fine-tuning pipeline from training through evaluation
- Designed Phase 6 Meeting Hall architecture: multiple specialist 7B models dividing hard problems by domain, solving autonomously, and combining solutions for vertical depth scaling without compute-intensive horizontal expansion

Chronos: LLM Time-Travel Debugger

(AWS EC2 • FastAPI • PostgreSQL • React Flow • Event Sourcing • SSE • Docker)

- Architected an event-sourced LLM observability platform that captures step-level traces (messages, tool calls/results, errors) and enables time-travel debugging: rewind to any state, fork "what-if" branches, and compare timelines to isolate divergence points
- Designed an interactive conversation DAG UI with live updates via server-sent events in React Flow, with inspectable failures and branch comparison via timeline playback and Arena metrics
- Deployed full stack on AWS EC2 using Docker Compose (backend, Postgres, Redis, pgAdmin) with health checks and restart policies; validated on AIMO3 mathematical reasoning benchmarks

Agent Persona Engine - Cognitive Architecture Extraction

(FastAPI • React • ChromaDB • Neo4j • OpenAI API • WebSockets)

- Solo-built cognitive architecture extraction engine creating synthetic expert personas from technical discussions, extracting 8 layers of cognitive DNA (identity, mental models, reflexes, reasoning chains, Big Five personality traits) for domain-specific performance over vanilla LLMs
- Engineered hybrid memory system combining ChromaDB vector search with Neo4j knowledge graph, achieving context-aware responses with sub-200ms retrieval and personality drift detection with autocorrection
- Built production React frontend + FastAPI backend with real-time WebSocket streaming, knowledge graph visualization, and side-by-side persona vs. vanilla GPT comparison

GestureFlow, Webcam Gesture Control System

(Python • MediaPipe • OpenCV • Landmark Classification)

- Built a real-time webcam gesture control system using MediaPipe landmark-based geometric ratios (scale-invariant, works at variable distances) with a five-gesture set for hands-free media control
- Engineered two-layer stability system: 5-frame rolling buffer + 0.7s hold timer, with 0.8s cooldown between actions to eliminate false positives in real-world usage

HACKATHONS & COMPETITIONS

- **ClawHack / MischiefClaw:** Built a satirical multi-agent escalation dashboard with dark UI, live Mission Control feed, and multi-agent cascade, single-file architecture with persistent state management
- **Kaggle – ‘Measuring Progress Toward AGI’:** Designed Metacognitive Depth Probe, a 60-question benchmark with ECE scoring and a fabricated-question tier as key differentiator for measuring AI reasoning depth
- **NVIDIA Nemotron Reasoning Challenge:** Built a meta-reasoning router using Reasoning Gym (104 task types) that classifies problems by type before selecting a solving strategy
- **Gemini Live Agent Challenge:** Shipped Fractal Form, an adaptive AI interviewer with branching goal trees, signal density scoring, and cross-session memory using Gemini 2.5 Flash + Live API

EDUCATION

Master of Science in Business Analytics and Project Management | University of Connecticut, Hartford, CT | Aug 2023 – May 2025

(Focus: Data Science & AI Systems | Relevant Coursework: Generative AI, Statistics, Data Science (Python), Big Data & Cloud, Predictive Modeling, Text Mining)

Bachelor of Engineering in Electronics and Communication | VNR VJIET, Hyderabad, India | Aug 2018 – Jul 2022

CERTIFICATIONS & RECOGNITION

AWS Certified AI Practitioner (Sep 2025, valid through Sep 2028) | IBM AI Foundations Coursework, 4-course series via Coursera (Aug 2020) | Active Open-Source Contributor | Hugging Face: SaiTejaMeka45 (published LoRA adapters, merged models, evaluation datasets)